

5. For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|x)$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|x) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t$, $t = 1, 2, \dots, T$. Note that $x = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - (\sum_{t=1}^T t)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.

$$\begin{aligned} \text{var}(b_2|x) &= \frac{\sigma^2}{\sum_{t=1}^T (t - \bar{t})^2} \\ \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{[T(T+1)/2]^2}{T} \\ &= \frac{T(T+1)}{2} \left(\frac{2T+1}{3} - \frac{T+1}{2} \right) = \frac{T(T+1)}{2} \times \frac{4T+2-3T-3}{6} = \frac{T(T+1)}{2} \times \frac{T-1}{6} \\ &= \frac{T(T^2-1)}{12} \\ \text{var}(b_2|x) &= \frac{12\sigma^2}{T(T^2-1)} \\ \lim_{T \rightarrow \infty} \text{var}(b_2|x) &= \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T^2-1)} = 0 \quad \therefore b_2 \text{ is consistent} \end{aligned}$$

b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t$, $t = 1, 2, \dots, T$. Here, $x = 0.5, 0.5^2, \dots, 0.5^T$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

$$\begin{aligned} y_t &= \beta_1 + \beta_2 (0.5)^t + e_t \\ x_t &= (0.5)^t \\ \text{var}(b_2|x) &= \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ \sum_{t=1}^T x_t &= \sum_{t=1}^T (0.5)^t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - (0.5)^T, \quad \bar{x} = \frac{1}{T} \sum_{t=1}^T x_t = \frac{1}{T} (1 - (0.5)^T) \\ \sum_{t=1}^T x_t^2 &= \sum_{t=1}^T (0.5)^{2t} = \frac{0.25(1 - 0.25^T)}{1 - 0.25} = \frac{1}{3} (1 - (0.25)^T) \\ \sum_{t=1}^T (x_t - \bar{x})^2 &= \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T} = \frac{1}{3} (1 - (0.25)^T) - \frac{(1 - (0.5)^T)^2}{T} \\ \text{var}(b_2|x) &= \frac{\sigma^2}{\frac{1}{3} (1 - (0.25)^T) - \frac{(1 - (0.5)^T)^2}{T}} \\ \lim_{T \rightarrow \infty} \text{var}(b_2|x) &= \lim_{T \rightarrow \infty} \frac{\sigma^2}{\frac{1}{3} (1 - 0) - \frac{(1 - 0)^2}{T}} = \frac{\sigma^2}{\frac{1}{3} \times 1 - 0} = 3\sigma^2 \neq 0 \quad \therefore b_2 \text{ not consistent} \end{aligned}$$

c. Provide an intuitive explanation for these results.

The difference in consistency between the two models comes from how the explanatory variable behaves as the sample size increases.

In part (a), the regressor $x_t = t$ increases over time, which leads to a growing spread in the data. As a result, $\sum (x_t - \bar{x})^2$ increases rapidly with T , causing the variance of the estimator to approach zero. This ensures that the estimator becomes more precise as more data are added, making it consistent.

In contrast, in part (b), the regressor $x_t = (0.5)^t$ converges quickly to zero. As T increases, new

observations provide very little additional variation, so $\sum (x_t - \bar{x})^2$ approaches a constant rather than increasing. Consequently, the variance of the estimator does not go to zero, and the estimator is not consistent.